

Figure 1: LP bias in SMALL samples, $T = 100, \rho = 0.95$ or $1, Nreps = 10,000$

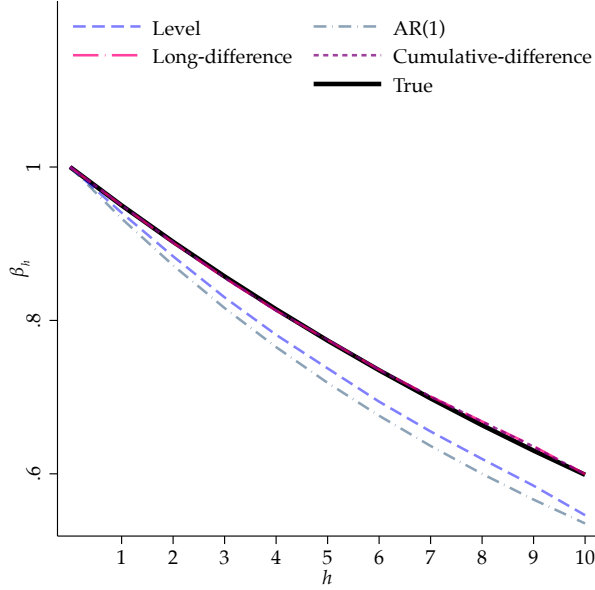
True DGP: $y_t = \rho y_{t-1} + u_t, \quad u \sim N(0, 1), \quad \rho = 0.95 \text{ or } 1, \quad T = 100$

Estimated OLS models:

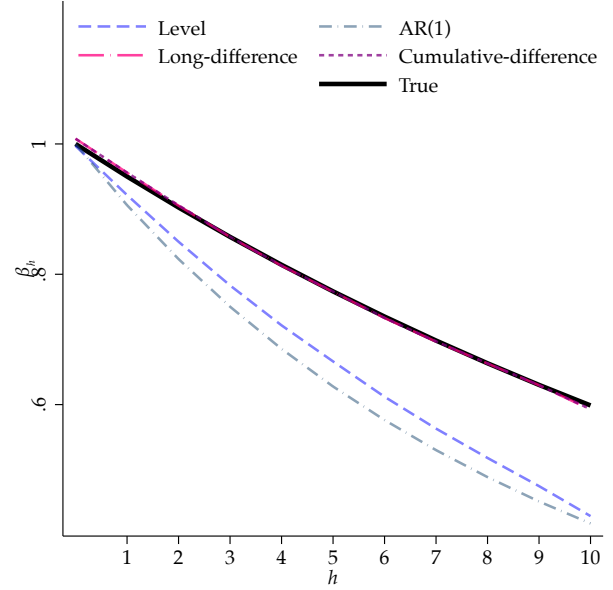
Intercept: no $y_t = \beta_1 y_{t-1} + u_t$

Intercept: yes $y_t = \beta_1 y_{t-1} + \beta_0 + u_t$

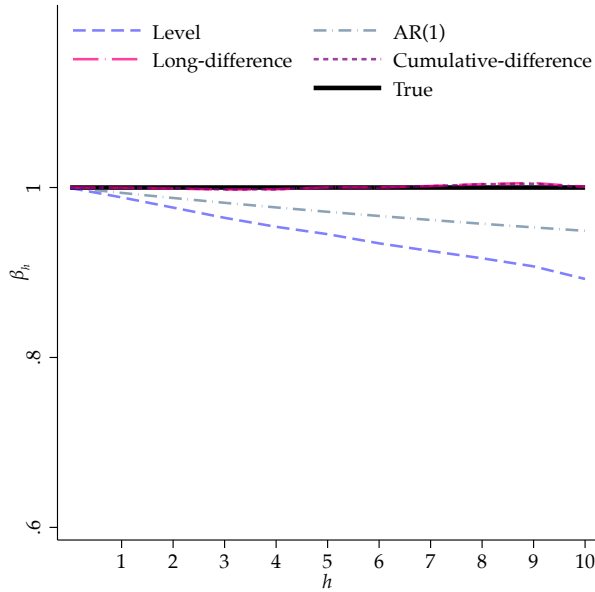
(a) $\rho = 0.95$; Intercept: no



(b) $\rho = 0.95$; Intercept: yes



(c) $\rho = 1$; Intercept: no



(d) $\rho = 1$; Intercept: yes

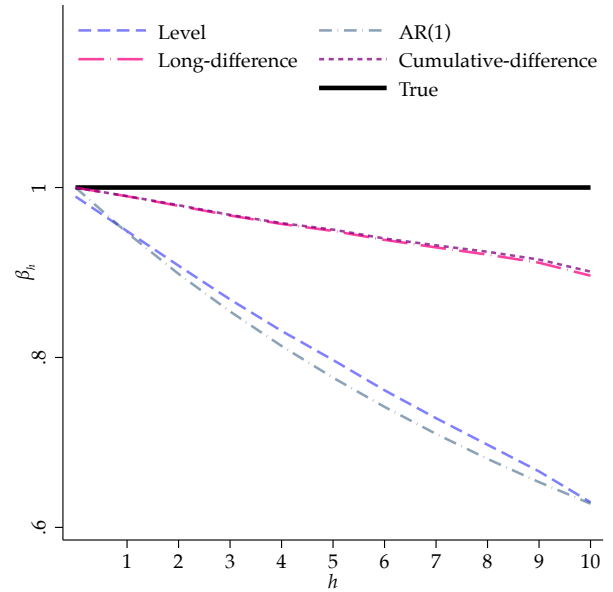


Figure 2: LP bias in LARGE samples, $T = 100,000, \rho = 0.95$ or $1, Nreps = 100$

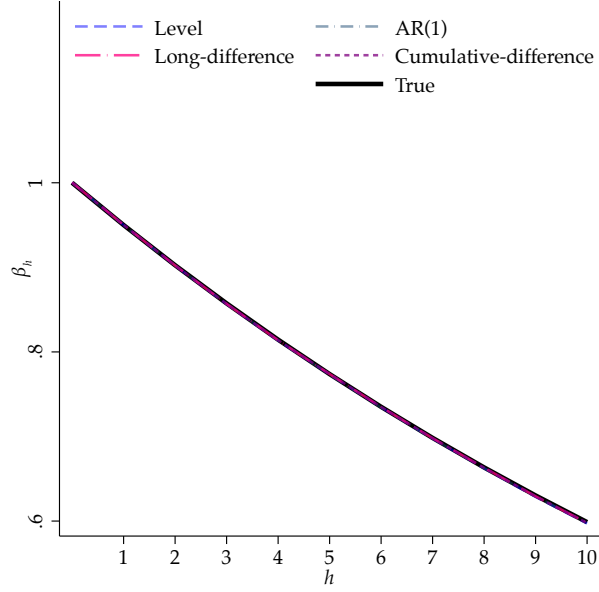
True DGP: $y_t = \rho y_{t-1} + u_t, \quad u \sim N(0, 1), \quad \rho = 0.95 \text{ or } 1, \quad T = 100,000$

Estimated OLS models:

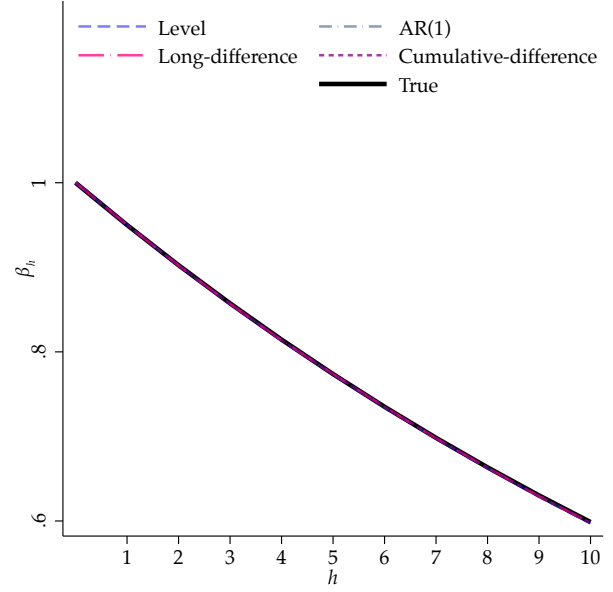
Intercept: no $y_t = \beta_1 y_{t-1} + u_t$

Intercept: yes $y_t = \beta_1 y_{t-1} + \beta_0 + u_t$

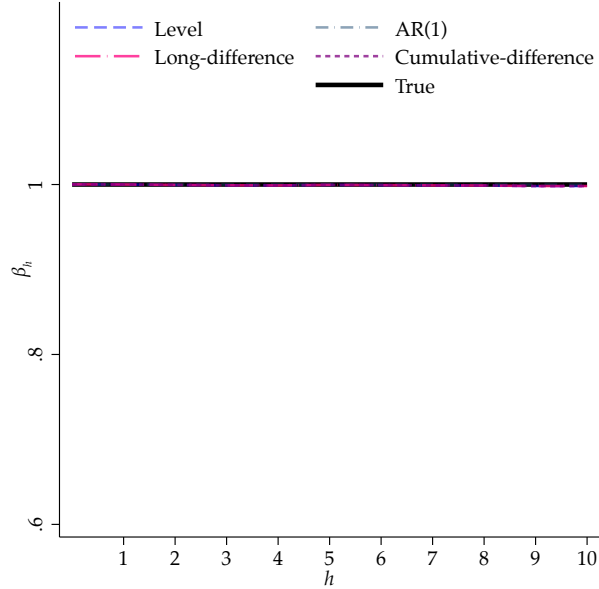
(a) $\rho = 0.95$; Intercept: no



(b) $\rho = 0.95$; Intercept: yes



(c) $\rho = 1$; Intercept: no



(d) $\rho = 1$; Intercept: yes

